Chinese Name_____ Student ID □□□□□□□□□□ School_____ Major_____ Sit NO._____

Sale Line ..... (DONOY WRITE YOUR ANSWER IN THIS AREA) ..... Sale Line ..... Sale Line .....

WARNING: MISBEHAVIOR AT EXAM TIME WILL LEAD TO SERIOUS CONSEQUENCE.

SCUT Final Exam

2020-2021-2《Calculus II》 Exam Paper A

Notice: **1. Make sure that you have filled the form on the left side of seal line.**
2. Write your answers on the exam paper.
3. This is a close-book exam.
4. The exam with full score of 100 points lasts 120 minutes.

Question No.	1-5	6-12	13-22	Sum
Score				

一. **Answer the questions.** $(3' \times 6 = 18')$

1. L is the boundary of circle $x^2 + y^2 = 9$ with counterclockwise direction, then curve integral $\oint_L (2xy - 2y)dx + (x^2 - 4x)dy =$ _____.

2. $L: x^2 + y^2 = a^2$ $\oint_L \sqrt{x^2 + y^2}\, ds =$ _____.

3. Let $D: x^2 + y^2 \le 2x$, write the $\iint_D f(x, y)dxdy$ in polar coordinate, then $\iint_D f(x, y)dxdy =$

4. If $\vec{a} = \{-1, 2, 2\}, \vec{b} = \{2, -1, 2\}$, then $(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) =$ _____.

5.The equation of plane through the three points $P_1(1, -2, 3), P_2(4, 1, -2), P_1(-2, -3, 0)$ is

_____.

6. The divergence of $\vec{A} = e^{xy}\vec{i} + \cos(xy)\vec{j} + xz^2\vec{k}$ =_____.

The divergence of $\vec{A} = e^{xy}\vec{i} + \cos(xy)\vec{j} + xz^2\vec{k}$

二、Finish the following questions. (7-16: $7'\times 10=70'$; 17-18: $6'\times 2=12'$)

7. Please show that $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{\sqrt{n}}$ is conditionally convergent.

8. Expand the function $f(x)=\arctan\frac{1+x}{1-x}$ into power series of x.

Solu:

9. Line integral $\int_L xy^2dx+y\varphi(x)dy$ is independent of path，and $\varphi(x)$ is derivative， $\varphi(0)=0$，find $\int_{(0,0)}^{(1,1)} xy^2dx+y\varphi(x)dy$.

10. Find $\int_{\frac{1}{4}}^{\frac{1}{2}}dx\int_{\frac{1}{2}}^{\sqrt{x}}e^{\frac{x}{y}}dy+\int_{\frac{1}{2}}^{1}dx\int_{x}^{\sqrt{x}}e^{\frac{x}{y}}dy$.

11. Find $\iiint_{\Omega}\left(x^2+y^2+z^2\right)dv$, Ω is bounded by $x^2+y^2+z^2=1$.

12. Find $\iiint_{\Omega} zdv$， and Ω is bounded by $x^2+y^2=1$ and $z=0$，$z=1$.

13. Let $f(u,v)$ is differentiable, $z=z(x,y)$is determined by $(x+1)z-y^2=x^2f(x-z,y)$ find $dz\big|_{(0,1)}$.

14. $\sum$ is the surface $z = x^2 + y^2 (z \le 1)$ with upper side, please calculate

$$I = \iint_{\sum} (x-1)^3 dydz + (y-1)^3 dzdx + (z-1)dxdy.$$

15. Find the minimum distance between the original point and the surface $z^2 = x^2 y + 4$..

Solu:

16. Let $z = f(u, x, y), u = xe^{y}, f$ has the second-order continuous partial derivative, find $\frac{\partial^2 z}{\partial x \partial y}$.

17. Find $\oint_L \frac{(x-1)dy - ydx}{(x-1)^2 + y^2}$, L represents a simple closed curve, including point $(1, 0)$ with

18. Find convergent region and sum function of power series $\sum_{n=0}^{\infty} \frac{x^{2n+2}}{(n+1)(2n+1)}$.